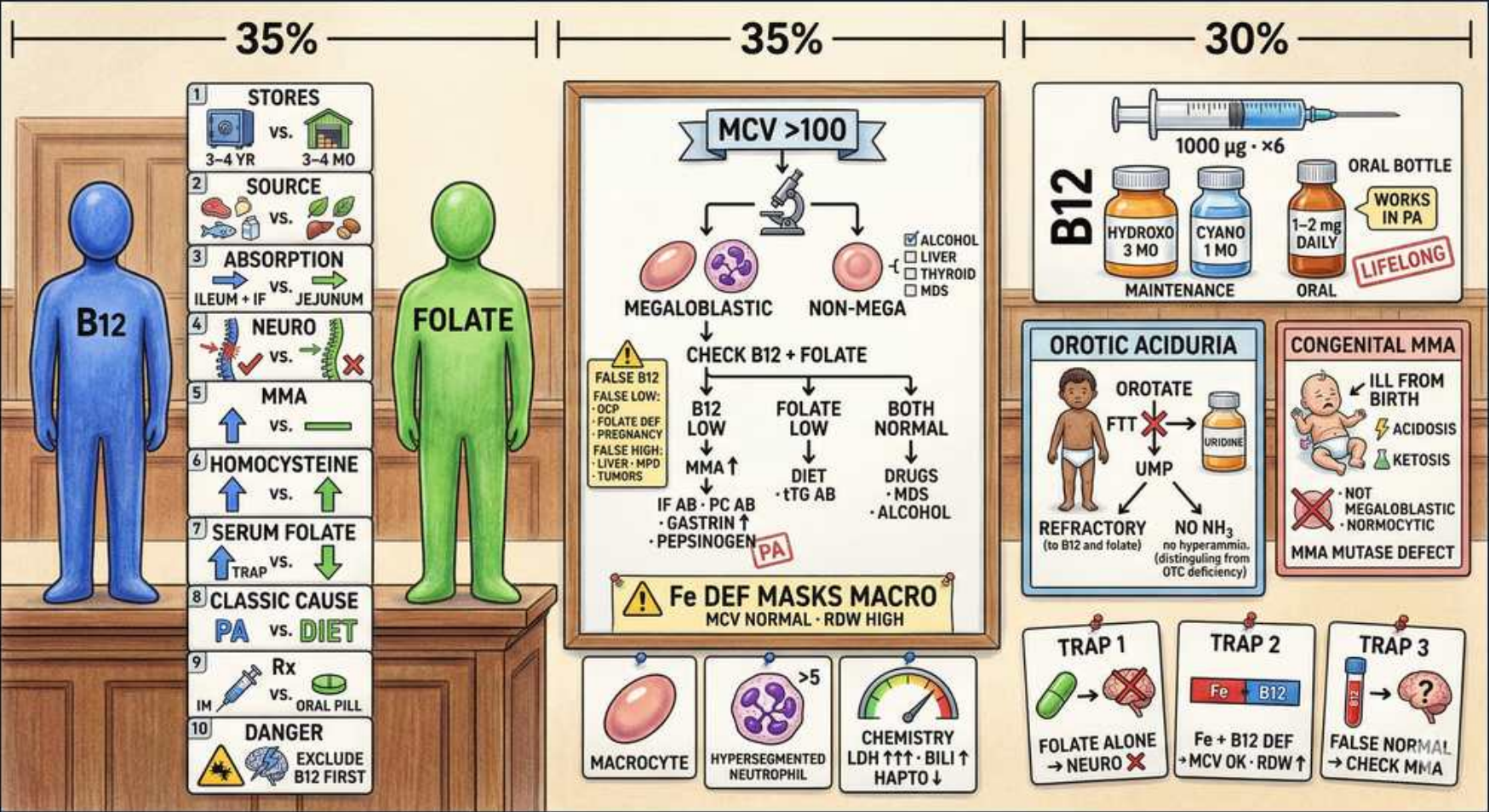


Megaloblastic — B12 vs Folate Comparison



1. Stores: 3-4 yr vs 3-4 mo

WHAT YOU SEE

'STORES 3-4 YR' (B12) vs '3-4 MO' (folate)

WHAT IT ENCODES

B12 hepatic stores last 3-4 YEARS (2-3 mg total, need ~2-5 µg/day); folate stores last only 3-4 MONTHS (~5-20 mg, need ~50-100 µg/day). So folate deficiency develops fast (months) while B12 deficiency is slow (years). Mnemonic: B12 = big vault (years); folate = small fridge (months).

BOARD TRAP

WRONG: expecting deficiency timelines to be similar. Discriminator: a megaloblastic anemia appearing within weeks-months of a new dietary/clinical insult (alcoholism, pregnancy, MTX) is far more likely FOLATE than B12 — B12 takes years.

2. Source: animal vs leaves

WHAT YOU SEE

'SOURCE meat/fish/dairy' vs 'leaves/liver/nuts'

WHAT IT ENCODES

B12 = animal products only (meat, fish, dairy, eggs); folate = green leafy vegetables, liver, legumes, nuts (heat-LABILE — destroyed by cooking). Mnemonic: B12 in animals, folate in foliage; boil the folate away.

BOARD TRAP

WRONG: assuming a heavy meat-eater can be folate deficient from diet alone or that a vegan rapidly drops folate. Discriminator: vegans risk B12 (animal source) but usually have ample folate from plants; the goat-milk-fed infant or 'tea-and-toast'/overcooked-diet elderly classically drop FOLATE.

3. Absorption: ileum+IF vs jejunum

WHAT YOU SEE

'ABSORPTION ILEUM + IF' vs 'JEJUNUM + PCFT'

WHAT IT ENCODES

B12 is absorbed in the TERMINAL ILEUM and requires intrinsic factor (+ cubilin receptor); folate is absorbed in the proximal JEJUNUM via PCFT after deconjugation to monoglutamate — no IF needed. Mnemonic: B12 = ileum + IF; folate = jejunum, just gets in.

BOARD TRAP

WRONG (favorite swap): saying B12 is absorbed in the jejunum or folate in the ileum. Discriminator: Crohn ileitis/ileal resection → B12 deficiency; celiac/tropical sprue/jejunal disease → folate deficiency (though severe diffuse disease can hit both).

4. Neuro: present vs intact

WHAT YOU SEE

'NEURO' damaged (B12) vs intact (folate)

WHAT IT ENCODES

B12 deficiency causes neurologic disease (subacute combined degeneration of dorsal columns + lateral corticospinal tracts, peripheral neuropathy, dementia/psychosis); folate deficiency does NOT cause neuro signs. This is THE cardinal discriminator. Mechanism: B12 also runs methylmalonyl-CoA mutase (myelin), independent of folate. Mnemonic: B12 protects the nerves; folate only fixes the marrow.

BOARD TRAP

WRONG (high-yield distractor): 'folate deficiency causes the same neurologic signs as B12.' It does NOT. Discriminator: new/worsening neuro signs with a macrocytic anemia point to B12 — and giving folate alone can let that neuro disease progress.

5. MMA: high vs normal

WHAT YOU SEE

'MMA' high (B12) vs normal (folate)

WHAT IT ENCODES

Methylmalonic acid (MMA) is HIGH in B12 deficiency (adenosyl-B12 runs methylmalonyl-CoA mutase) but NORMAL in folate deficiency (folate is not in that pathway). MMA is the specific biochemical discriminator and is useful for borderline B12 levels. Mnemonic: MMA = B12's fingerprint.

BOARD TRAP

WRONG: 'folate deficiency raises MMA.' It does NOT. Discriminator: high MMA + high homocysteine = B12; normal MMA + high homocysteine = folate. MMA is the test that separates the two.

6. Homocysteine high in both

WHAT YOU SEE

'HOMOCYSTEINE' up in both

WHAT IT ENCODES

Homocysteine is elevated in BOTH B12 and folate deficiency because both are cofactors for methionine synthase (homocysteine → methionine). It is sensitive for deficiency but cannot distinguish which vitamin. Mnemonic: homocysteine = shared road; MMA = the fork that tells them apart.

BOARD TRAP

WRONG: using homocysteine to decide between B12 and folate. Discriminator: homocysteine rises in both — pair it with MMA (high only in B12) to separate them. Isolated high homocysteine with normal MMA favors folate (or B6/MTHFR issues).

7. Serum folate: trap vs down

WHAT YOU SEE

'SERUM FOLATE trap↑' (B12) vs 'down' (folate)

WHAT IT ENCODES

The METHYLFOLATE TRAP: in B12 deficiency, methionine synthase fails, so folate gets stuck as methyl-THF and cannot regenerate usable THF — serum folate may appear normal/HIGH even though tissue folate is functionally low. In true folate deficiency, serum folate is LOW. Mnemonic: low B12 traps folate as methyl-THF → serum folate paradoxically up.

BOARD TRAP

WRONG: interpreting a normal/high serum folate as 'folate is fine, this can't be deficiency.' Discriminator: the methylfolate trap from B12 deficiency can elevate serum folate; check B12/MMA and RBC folate rather than trusting serum folate alone.

8. RBC folate down in both

WHAT YOU SEE

'RED CELL FOLATE down 2/3' both

WHAT IT ENCODES

RBC (red-cell) folate reflects tissue folate over the prior ~120 days and falls in BOTH folate AND B12 deficiency (because the methylfolate trap depletes usable tissue folate). It is more reliable than serum folate but is not specific for which vitamin. Mnemonic: RBC folate = tissue truth, but it drops in both.

BOARD TRAP

WRONG: using a low RBC folate to confirm folate deficiency specifically. Discriminator: RBC folate is low in both deficiencies, so a low value cannot exclude B12 — confirm B12 status with B12 level + MMA before labeling it folate deficiency.

9. Classic cause: PA vs diet

WHAT YOU SEE

'CLASSIC CAUSE PA' vs 'DIET/PREGNANCY'

WHAT IT ENCODES

Classic B12 cause = pernicious anemia (autoimmune loss of intrinsic factor/parietal cells); also vegan diet, ileal disease, metformin, PPIs. Classic folate causes = poor diet/alcoholism, pregnancy/lactation, methotrexate/trimethoprim/phenytoin. Mnemonic: B12 = PA (Pernicious Anemia); folate = Pregnancy + Poor diet + Pills (antifolates).

BOARD TRAP

WRONG: attributing every macrocytic anemia in an alcoholic to B12. Discriminator: alcoholism classically causes FOLATE deficiency (poor intake + impaired metabolism); think B12 when there are neuro signs, pernicious-anemia antibodies, ileal disease, or metformin use.

10. Rx: IM syringe vs oral pill

WHAT YOU SEE

'Rx IM SYRINGE' vs 'ORAL PILL'

WHAT IT ENCODES

B12 is classically replaced parenterally (IM hydroxocobalamin/cyanocobalamin — e.g., 1000 µg loading then maintenance), essential when IF is absent (PA); high-dose oral 1000-2000 µg/day works via passive diffusion. Folate is replaced orally (1-5 mg/day). Mnemonic: B12 often needs a shot (no IF); folate is a simple pill.

BOARD TRAP

WRONG: giving standard low-dose oral B12 to a pernicious-anemia patient. Discriminator: without intrinsic factor, normal oral doses aren't absorbed — use IM (or high-dose oral by passive diffusion). And never give folate alone for macrocytic anemia until B12 is excluded.

11. MCV >100 → check both

WHAT YOU SEE

CENTER 'MCV >100' → megaloblastic vs non-mega → check B12 + folate

WHAT IT ENCODES

MCV >100 fL = macrocytosis. Split into MEGALOBLASTIC (B12/folate deficiency, antimetabolite drugs, orotic aciduria → hypersegmented neutrophils) vs NON-megaloblastic (alcohol, liver disease, hypothyroidism, MDS, reticulocytosis — no hypersegmentation). Then send B12 + folate. Mnemonic: macrocytosis splits mega (impaired DNA) vs non-mega (membrane/marrow).

BOARD TRAP

WRONG: assuming all macrocytosis is B12/folate deficiency. Discriminator: alcohol, liver disease, hypothyroidism, and MDS cause non-megaloblastic macrocytosis with NORMAL B12/folate and NO hypersegmented neutrophils — look for hypersegmentation + check the vitamins before treating.

12. Fe def masks macro

WHAT YOU SEE

'Fe DEF MASKS MACRO', 'MCV NORMAL RDW HIGH'

WHAT IT ENCODES

Coexisting iron deficiency (microcytic) can cancel out the macrocytosis of B12/folate deficiency, producing a NORMAL MCV but a HIGH RDW (dimorphic smear with both micro- and macrocytes). Mnemonic: micro + macro = normal MCV, but RDW gives it away.

BOARD TRAP

WRONG: 'a normal MCV rules out megaloblastic anemia.' It does NOT. Discriminator: a high RDW, hypersegmented neutrophils, or anemia despite normal MCV should prompt iron studies + B12/folate — combined deficiencies hide behind a normal MCV.

13. Trap: folate alone → neuro

WHAT YOU SEE

TRAP 1 'FOLATE ALONE in B12 def → ANEMIA OK NEURO worse'

WHAT IT ENCODES

Giving folate alone to a B12-deficient patient corrects the megaloblastic ANEMIA (folate restarts DNA synthesis) but does NOT fix — and can WORSEN/unmask — the B12 neurologic disease, which progresses silently. Always exclude/replace B12 first. Mnemonic: folate fixes the blood and hides the danger to the nerves.

BOARD TRAP

WRONG: empirically treating macrocytic anemia with folate. Discriminator: check B12 first; folate monotherapy in occult B12 deficiency can precipitate or accelerate subacute combined degeneration while the CBC looks better.

14. Trap: false-normal B12 in PA

WHAT YOU SEE

TRAP 3 'FALSE NORMAL B12 in PA → check MMA'

WHAT IT ENCODES

Serum B12 can read falsely NORMAL even in true deficiency/pernicious anemia (most B12 is bound to inert haptocorrin; assay limitations). When clinical suspicion is high or the level is borderline (~150-300 ng/L), confirm with MMA + homocysteine (both high = B12). Mnemonic: don't trust a borderline B12 — let MMA be the judge.

BOARD TRAP

WRONG: excluding B12 deficiency on the basis of a 'normal' serum B12 in a patient with neuro signs or PA antibodies. Discriminator: send MMA (and homocysteine) and/or holotranscobalamin; an elevated MMA confirms functional B12 deficiency despite a normal total B12.

15. Orotic aciduria (folate/B12-normal)

WHAT YOU SEE

OROTIC ACIDURIA panel: orotate→UMP block, '↑orotate in urine', refractory to B12/folate, NO hyperammonemia

WHAT IT ENCODES

Hereditary orotic aciduria is a rare megaloblastic anemia from a defect in UMP synthase (orotate → UMP) in pyrimidine synthesis. It presents with megaloblastic anemia + failure to thrive + high urinary OROTIC ACID and is REFRACTORY to B12 and folate; treatment is oral URIDINE (bypasses the block). Mnemonic: orotic aciduria = megaloblastic anemia that ignores B12/folate, fixed by uridine.

BOARD TRAP

WRONG: confusing it with ornithine transcarbamylase (OTC) deficiency, which ALSO raises orotic acid. Discriminator: hereditary orotic aciduria has NO hyperammonemia and causes megaloblastic anemia; OTC deficiency causes HYPERAMMONEMIA (encephalopathy) with NO megaloblastic anemia.

16. Congenital MMA (mutase defect)

WHAT YOU SEE

CONGENITAL MMA panel: 'ill from birth', acidosis/ketosis, NOT megaloblastic/normocytic, MMA MUTASE defect

WHAT IT ENCODES

Congenital methylmalonic acidemia results from an inherited defect in methylmalonyl-CoA mutase (or adenosyl-B12 synthesis), causing high MMA WITHOUT B12 deficiency. Neonates present early with metabolic acidosis, ketosis, and failure to thrive; the anemia is normocytic, NOT megaloblastic. Mnemonic: high MMA but normal B12 and normal MCV from birth = inherited mutase defect.

BOARD TRAP

WRONG: interpreting a high MMA as proof of dietary/acquired B12 deficiency. Discriminator: in congenital MMA the B12 level is NORMAL and the picture is a sick acidotic infant — confirm B12 status; high MMA only means B12 deficiency when intake/absorption is actually impaired (some forms respond to high-dose B12).
